

Data Visualisation and Insight

Group PROJECT – Dashboard using array of technologies

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| **Tableau:** Marek | **Power BI:** Santhosh |

Summary

The Accident Insights dashboards serve as an interactive visual data tool which enables users to investigate road accident patterns and the factors which cause them. There are two primary aims. One is to compare current year and previous year accident data to identify trends in accident frequency, fatalities, and severity. The other is to find out the cause and number of deaths in a certain scenario.

The dashboard, made in Power BI, displays current year accident data by urban and rural regions while it shows monthly fatality and accident data changes between the present year and past year. The visualizations study accidents according to road conditions which include icy roads snow-covered roads wet roads and dry roads and road types which include main roads streets and highways. The special section presents the most common accident causes which include distracted driving drunk driving mechanical failure speeding and weather. Accidents are examined according to their month of occurrence and day of the week and time of day to discover temporal patterns.

The dashboard created in Tableau focused on displaying different scenarios and what poses a greater risk. The heat map visualization shows the severity of accidents by cause of accident according to selected metrics. The line chart of fatalities, vehicles involved, and injuries shows the number depending on speed. The severity distribution based on the road condition chart calculates severity and visually displays the result. The user can select aspects to see different scenarios, allowing them to visualize and understand the possibilities of accidents and their potential consequences.

For both dashboards, users can interactively filter the data and see various results. Dashboards enable road safety analysts, transport authorities, policymakers, and researchers to make decisions based on data analysis.

Data Source and Description

The Global Road Accidents Dataset from Kaggle serves as the dataset for this research work. The dataset includes information about road traffic accidents together with multiple factors which may affect the probability of accidents and the severity of those accidents. It provides details about the conditions which existed during accidents together with the environmental elements and driver attributes.

The data file underwent multiple Python data cleaning procedures which prepared it for analytical processing. The data was loaded into a Pandas DataFrame using the `read_csv()` function. The structure of the dataset was evaluated using `df.shape`, `df.info()`, and `df.describe()`,

which provided information about the record count, data types, and summary statistics. The dataset analyzed all missing information through `isnull().sum()` which was used to find all incomplete records. The dataset underwent duplicate testing through `duplicated().sum()` to ensure that no repeated records existed, which would compromise the analysis. The existing format of the dataset enabled us to perform exploratory analysis and visualisation work without needing any structural changes to the dataset.

For the cleaned dataset. We used several exploratory data analyses (EDA) to identify key elements that we should use to tell the story to users. We mainly used bar charts to identify these elements, but also heatmap and pie chart. The actual result can be seen in the python file, where detailed results are showed. In short, our findings let to realization we have synthetic or non-telling dataset. There was a discussion of what should be the next step to do. The decision was made, and we continued with the same data set. For such a dataset, we decided to split workload according to specific topics/themes.

Marek's theme: "Why Are Accidents Severe And What It Can Cause It?"

Santhosh's theme: "Where and When Do Accidents Happen?"

Techniques and Features

Two dashboards meant to be created be tools such a PowerBI and Tableau. Themes/topics were split across two members of the group. Each dashboard should show intractability, visualisation and well-designed layout.

Tableau:

- **Line chart:** This graph illustrates the impact of speed limits. It shows the effect of speeds ranging from 30 km/h to 120 km/h. It allows you to visualize the consequences of high speeds.
- **Heatmap:** Likely the most complex graph. It visualize the Accident Cause, Accident Severity and select specific measure (Cyclists Involved, Driver Alcohol level, Driver Fatigue, Visibility Level) to see the impact and the differences.
- **Tree map:** It is used as a visually appealing pie chart. Shows the consequences of driving on specific type of road.
- **Bar chart:** A graph that tells us the story of the influence of population density on accident severity. It should show us how population density changes based on accidents that occur in the surrounding area

Power BI:

- **KPI Cards:** The KPI cards displayed essential metrics which included Total Accidents and Total Casualties and Severe Accidents and Moderate Accidents. The selection of KPI cards enables users to get instant access to critical metrics which display all main performance indicators through brief summaries.
- **Donut Chart:** The donut chart displayed accident distribution between urban and rural areas. This visualisation method was selected because it shows proportionate distribution across categories which enables direct comparison of accident distribution between two location categories through visualisation instead of table data.

- **Line Chart:** The line chart displayed monthly comparison of current year fatalities against previous year fatalities. Line charts allow users to track time-based trends because they display data from multiple months which enables users to see both upward and downward patterns.
- **Clustered Bar Chart:** The clustered bar chart displayed accident data from different road conditions which included icy roads and snow-covered roads and wet roads and dry roads. The chart was selected because it enables users to compare two variables which include current year data and previous year data across different categories.
- **Clustered Column Chart:** The clustered column chart displays accident data for different road types which include main roads and streets and highways. The visualisation was selected because column charts show better value comparison between categories which helps viewers identify the existing differences.

Individual Contribution

The project required me to complete all work by myself. I was responsible for the entire data visualization process, including:

- Collecting and importing the dataset
- Preparing the dataset for analysis
- Designing the dashboard layout
- Creating all charts and visualisations in Power BI
- Implementing interactive filters and navigation elements
- Analysing accident patterns and trends
- Preparing the final report and documentation

Appraisal (Individual)

Findings from the Group Appraisal:

My team examined the dashboard system to determine its capability to display road accident information. The evaluation measured three different aspects through visual display elements which showed dashboard functionality and matching design elements and their ability to show information clearly.

Strengths of the Dashboard:

The review process discovered various positive elements which existed throughout the assessment of the dashboard system.

Clear KPI Indicators:

The dashboard contains well-constructed KPI cards which show essential performance indicators for total accidents and total casualties and severe accidents and moderate accidents.

Effective Representation of Percentage Changes:

The implementation of percentage change indicators together with KPI values enables users to track development patterns from earlier time frames

Consistent Color Usage for Year Comparison:

The dashboard uses consistent colors to show both the current year and the previous year throughout its different visualizations. Enhancement allows users to understand content better because they can evaluate values through various chart displays.

Clear Labelling of Accident Causes:

The accident cause section presents its categories through distinct labels which include distracted driving and drunk driving and mechanical failure and speeding and weather-related accidents. The labels and numerical values make it easy for users to quickly identify the most common causes of accidents.

Effective Use of Clustered Charts for Comparison:

The proper application of clustered charts which display data together with their independent elements allows for effective comparison. The clustered bar and column charts enable users to compare accident data which they gather from multiple road conditions and road type categories.

Areas for Improvement:

The dashboard provides valuable insights yet there exist multiple areas which need enhancement.

Incorrect Month Sorting on X-Axis:

Some charts display months on their X-axis which do not show the proper chronological sequence. Users will find this information confusing which will result in less precise trend assessment

Inconsistent Color Usage Across Some Visualizations:

Some charts use the same color scheme to display different years while other Visualizations choose different color patterns. Visual clarity would benefit from using a unified color scheme throughout all charts.

Overuse of Donut Charts:

The dashboard loses its effectiveness because users include excessive donut charts which functions to show proportions. The visualization should use other methods which include bar charts and stacked charts to show better comparison results.

Limited Context for Some Visualizations:

Some graphs present numerical values without additional context or explanations. The understanding can be enhanced through the addition of tooltips and annotations and descriptive titles

Recommendations for Future Work

The dashboard requires multiple improvements which designers should implement during their upcoming development work.

Improve Data Sorting and Formatting:

Correct month sorting will enhance time-based data display accuracy which helps users understand trend visualizations better.

Maintain Consistent Colour Schemes:

The use of standardized color palettes across all visualizations will create better visual uniformity which enables users to conduct easier comparisons.

Use Additional Visualisation Types:

Future dashboards could incorporate additional visualization techniques such as:

- Heatmaps which display accident hotspot locations
- Geographic maps that show accident distribution across different areas
- Stacked bar charts which show different categories for comparison purposes
- These visualizations will show more detailed information about accident patterns.

